

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT _*CO-SC(MODEL-A&APEX) | Date: 15-03-26
Time: 3 Hrs | Max. Marks: 180

15-03-26_SR.IIT_STAR CO-SC(MODEL-A&APEX)_JEE ADV_RTA-6_SYLLABUS

MATHEMATICS: **PRESENT WEEK (100%):** Logarithms & Quadratic equation, Sequence and series, Binomial Theorem, Trigonometric Functions

PHYSICS: **PRESENT WEEK (100%):** Heat Transfer, Fluid Statics & Dynamics, Elasticity & Viscosity, Surface Tension, Units-Measurements

CHEMISTRY: **PRESENT WEEK (100%):** Equilibrium, Hydrocarbon, Benzene, Chemical Bonding

Rankers Academy JEE
THE NARAYANA GROUP

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TIME:3hrs **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

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MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 4)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 5 – 7)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – III (Q.N : 8 – 13)	Questions with **Non-Negative Integer** type	+4	0	6	24
Sec – IV (Q.N : 14 – 17)	Questions with **Comprehension Type** With **Numerical value type** (3 Comprehensions – 2 + 2 = 4Q)	+3	0	4	12
Total				**17**	**60**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 18 – 21)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 22 – 24)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – III (Q.N : 25 – 30)	Questions with **Non-Negative Integer** type	+4	0	6	24
Sec – IV (Q.N : 31 – 34)	Questions with **Comprehension Type** With **Numerical value type** (3 Comprehensions – 2 + 2 = 4Q)	+3	0	4	12
Total				**17**	**60**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 35 – 38)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 39 – 41)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – III (Q.N : 42 – 47)	Questions with **Non-Negative Integer** type	+4	0	6	24
Sec – IV (Q.N : 48 –51)	Questions with **Comprehension Type** With **Numerical value type** (3 Comprehensions – 2 + 2 = 4Q)	+3	0	4	12
Total				**17**	**60**

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MATHEMATICS

MAX.MARKS: 60

SECTION–I (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

1. If $n=\binom{40}{0}\binom{20}{10}+\binom{41}{1}\binom{19}{9}+\binom{42}{2}\binom{18}{8}+\cdots+\binom{50}{10}\binom{10}{0}$, then the number of trailing zeros at the end of n is:

A) 0 B) 1 C) 2 D) 3

2. The value of $\sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\sum_{p=1}^{\infty}\frac{m^2n^2p}{2^{m+n}(mn2^p+mp2^n+np2^m)}$

A) 2 B) 5/3 C) 1 D) 8/3

3. How many polynomials of the form $P(x)=x^3+ax^2+bx+c$ having zeros are a, b, c (a, b, c $\in$ R)

A) 2 B) 3 C) 0 D) 1

4. The value of $2^{20}-{}^{19}C_1 2^{18}+{}^{18}C_2 2^{16}-{}^{17}C_3 2^{14}+{}^{16}C_4 2^{12}\ldots.+{}^{10}C_{10}2^0$

A) 0 B) 1 C) 20 D) 21

SECTION - II (Maximum Marks : 12)

This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

5. The complete set of values of real parameter 'm' for which the equation $m\sin^2 x+(m-1)\sin x+(m-2)=0$ has

A) No real roots is $(-\infty,1)\cup\left(1+\frac{2}{\sqrt{3}},\infty\right)$ B) 5 roots in $[0, 2\pi]$ is $\{2\}$

C) 4 roots in $[0, 2\pi]$ is $(1,2)\cup\left(2,\frac{\sqrt{3}+2}{\sqrt{3}}\right)$ D) 2 roots in $[0, 2\pi]$ is $\left\{-1,1,\frac{\sqrt{3}+2}{\sqrt{3}}\right\}$

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6. Given that a_1, a_2, a_3 is an A.P in that order, $a_1+a_2+a_3=15$; b_1, b_2, b_3 is a G.P. in that order, and $b_1b_2b_3 = 27$. If $a_1 + b_1$, $a_2 + b_2$, $a_3 + b_3$ are positive integers and form a G.P. in that order, then

A) maximum possible value of a_3 is 15(a_1, a_2, a_3 are positive integer and unequal)

B) minimum possible value of a_3 is 3(a_1, a_2, a_3 are positive integer and unequal)

C) maximum possible value of a_3 is $9+\frac{55+7\sqrt{61}}{2}$

D) maximum possible value of a_3 is $4+\frac{55+7\sqrt{61}}{2}$

7. If $\sum_{r=0}^{9} {}^{20}C_{2r}\ {}^{20}C_{2r+2} = \frac{a}{a+1}\left({}^{39}C_{21} - {}^{19}C_{10}\right)$, then which of the following options is/are correct

A) a = 10

B) a = 9

C) The total number of ordered pair solution (x, y) where $x, y \in I$ for $xy = 4a$ is 16

D) The total number of ordered pair solution (x, y) where $x, y \in I^+$ for $xy = 4a$ is 8

SECTION-III (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

8. If $a_1, a_2, a_3, \ldots$ are in H.P, $a_1 > 0$ for each i, then: $\sum_{i=1}^{n} \frac{n(a_i)^{2/3}(a_{i+1})^{2/3}}{(a_i)^{2/3}+(a_{i+1})^{2/3}+(a_i)^{1/3}(a_{i+1})^{1/3}}$ is $\frac{an^2}{b}$

Where a and b are coprime natural number then b – a is (where $a_1 = 1$, $a_{n+1} = 8$).

9. If a, b, c are real numbers such that $3\left(a^2+b^2+c^2+1\right) = 2(a+b+c+ab+bc+ca)$ then the value of $\frac{3a+4b+5c}{3b+a}$

10. Let a, b, c, d be the roots of $x^4 - x^3 - x^2 - 1 = 0$. Also consider $g(x) = x^6 - x^5 - x^3 - x^2 - x$. Then the value of g(a) + g(b) + g(c) + g(d)

11. If $\frac{\sum_{r=1}^{99}\sqrt{10-\sqrt{r}}}{\sum_{r=1}^{99}\sqrt{10+\sqrt{r}}} = a\sqrt{b} - c$ then value of a + b + c ($\because$ 'b' be a prime number)

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12. If x_1, x_2, x_3, x_4 are non-zero real numbers such that

$$x_1 - \frac{1}{x_1} = 2x_2$$

$$x_2 - \frac{1}{x_2} = 2x_3$$

$$x_3 - \frac{1}{x_3} = 2x_4$$

$$x_4 - \frac{1}{x_4} = 2x_1;$$

then number of 4 tupples (x_1, x_2, x_3, x_4) satisfying the above equation.

13. If $a^2 + ab = 12$, $2ab + 3b^2 + 5 = 0$ and $\frac{a}{9} \neq \frac{-b}{5}$ then a + 4b is equal to

SECTION - IV (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Paragraph-I for Question Nos. 14 and 15:

Let $A = \sum_{r=1}^{30} \frac{(30+r)!(30-r)!}{[(30)!]^2}\left(\frac{2r-1}{30+r}\right)$

$B = \sum_{r=0}^{30} \left({}^{30}c_r\right)^2$

$C = \sum_{r=0}^{30} (-1)^r \left({}^{60}c_r\right)^2$

14. The value of C – A is

15. The value of (B + C – 2A) is

Paragraph-II for Question Nos. 16 and 17:

Let $P(x) = x^5 - 9x^4 + px^3 - 27x^2 + qx + r (p, q, r \in R)$ be divisible by x^2 and α, β and γ are the positive roots of the equation $\frac{P(x)}{x^2} = 0$.

16. The value of (p + q + r) is equal to

17. If $\alpha - 1, \beta + 3$ and $\gamma + 7$ are the first three terms of a sequence whose sum of first n terms is given by S_n then $\sum_{n=2}^{\infty} \frac{1}{\sqrt{S_n \cdot S_{n-1}}}$ is equal to

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PHYSICS **MAX.MARKS: 60**

SECTION–I (Maximum Marks: 12)

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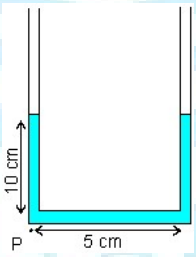
18. A flywheel can rotate in order to store kinetic energy. The flywheel is a uniform disc made of a material with a density ρ and tensile strength σ (measured in Pascals), radius r, and thickness h. The flywheel is rotating at the maximum possible angular velocity so that it does not break. Which of the following expression correctly gives the maximum kinetic energy per kilogram that can be stored in the fly wheel ? Assume that α is a dimensionless constant.

A) $\alpha\dfrac{\sigma}{\rho}$ B) $\alpha h\sqrt{\dfrac{\rho\sigma}{r}}$ C) $\sigma\alpha\sqrt{\dfrac{h\rho}{r}}$ D) $\alpha\dfrac{h\sigma}{r\rho}$

19. A thin circular disc of mass m, uniform surface charge density σ and having radius R is placed on a thin film of viscous liquid of thickness t and coefficient of viscosity η. The film has circular cross-section of same area as that of the disc. A uniform magnetic field exists perpendicular to the plane of the disc and in same area. If the magnetic field starts increasing at a constant rate β, what angular velocity (about its axis) must be given to the disc so that it continues to rotate with the same angular velocity ?

A) $\dfrac{\sigma\beta t}{2\eta}$ B) $\dfrac{2\sigma\beta t}{3\eta}$ C) $\dfrac{3\sigma\beta t}{2\eta}$ D) $\dfrac{3\sigma\beta t}{4\eta}$

20. When a thin U-shaped tube is placed near surface of earth, liquid level is same on both vertical arms of the tube as shown. Now a particle 3 times mass of earth but of negligible size is placed below left arm at point P. Then liquid level in left arm will be



A) 5.625 cm B) 10.625 cm C) 9.375 cm D) 8 cm

21. A nuclear weapon can be assumed as a spherical ball of radius $\frac{1}{2}$ m. Surface of weapon can be treated as blackbody with surface temperature of 10^6 K at detonation. If all energy is in the form of radiation then how much momentum does it impart per unit area per unit time on a perfect mirror at 1 km distance? Assume normal incidence on mirror and only direct radiation. $\sigma=\dfrac{17}{3}\times10^{-8}\ \text{W/m}^2\text{K}^4$

A) 94.4 B) 47.7 C) 23.85 D) 188.8

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SECTION - II (Maximum Marks : 12)

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Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

22. 'Gulab Jamuns' (assumed to be spherical) are to be heated in an oven. They are available in two sizes, one twice bigger (in radius) than the other. Pizzas (assumed to be discs) are also to be heated in oven. They are also in two sizes, one twice big (in radius) than the other. All four are put together to be heated to oven temperature. Choose the correct option from the following:

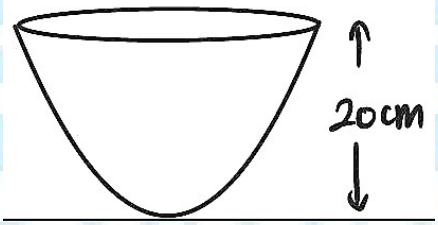
A) Both size gulab jamuns will get heated in the same time.

B) Smaller gulab jamuns are heated before bigger ones.

C) Smaller pizzas are heated before bigger ones.

D) Bigger pizzas are heated before smaller ones.

23. A paraboloid is made by rotating a parabola $y = 2x^2$. This uniform solid paraboloid object has density σ. Height of paraboloid is h = 20 m. It is now released from a position so that lower end just touches liquid of density ρ. It again comes to rest when it just submerges completely in liquid. Then (Take $3^{1/4} = 0.75$)



A) Ratio of densities of liquid and object is 3:1

B) Ratio of densities of liquid and object is 3:2

C) Loss in mechanical energy when object finally stops oscillating at equilibrium is $\frac{\sigma\pi g h^3}{6\sqrt{3}}$

D) Angular frequency of small oscillations about equilibrium position is $\sqrt{\frac{2\sqrt{3}g}{h}}$

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24. A sphere of radius R is maintained at 600 K in surrounding temperature of 300 K. Sphere can be considered black body. A power source inside sphere provides energy to keep it's temperature constant. Another similar sphere of radius R′ (Also blackbody) laminated with a foil (e = 1) is maintained at 600 K using source of same power as previous one when kept in same surrounding.

A) Ratio of $\frac{R'}{R} = \sqrt{2}$ B) Ratio of $\frac{R'}{R} = \sqrt{\sqrt{2}}$

C) Temperature of lamination will be nearly 512 K

D) Temperature of lamination will be nearly 488 K

SECTION-III (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**

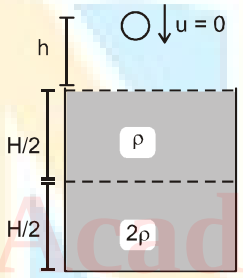
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.

Answer to each question will be evaluated according to the following marking scheme:

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Zero Marks: 0 In all other cases.

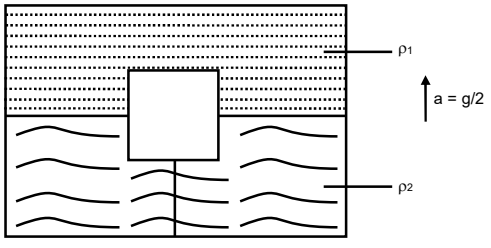
25. A small ball of mass m is released from rest from a height h above the liquid surface. The density of ball is $\frac{\rho}{3}$: (assume no splash)(H = 24 cm)



What should be the minimum value of height of the ball (in cm) from bottom of the container so that ball just reaches the bottom of the vessel?

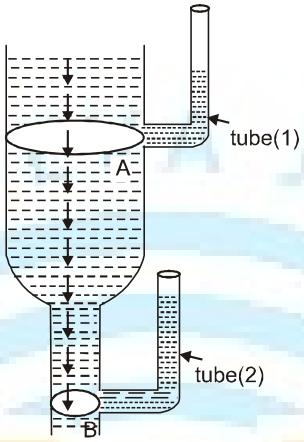
26. A vertical capillary tube with inner radius 0.5 mm, is submerged into water so that the length of its part above the water surface is h = 25 mm. Radius of curvature of meniscus formed is $\frac{\alpha}{10}$ mm. Here α is an integer. Find 2α. (surface tension of water = 0.075 N/m. Density of water = 10^3 kg/m^3, angle of contact = 0º)

27. A vessel contains two immiscible liquids of densities $\rho_1 = 1000\,\text{kg m}^{-3}$ and $\rho_2 = 1500\,\text{kg m}^{-3}$. A solid block of volume $V = 10^{-3}\,\text{m}^3$ and density d = 800 kg m^{-3} is tied to one end of a string and other is tied to the bottom of the vessel as shown in figure. The block is immersed with $\frac{2^{th}}{5}$ of its volume in the liquid of higher density and $\frac{3^{th}}{5}$ in the liquid of lower density. The entire system is kept in an elevator which is moving upwards with an acceleration of a = g/2. Find the tension in the string. (Take g = 10 m s^{-2}) (in N)



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28. A non–viscous ideal fluid is flowing vertically downwards in a pipe. The area of cross-section at section 'A' is double that at section 'B'. Vertical distance between section A and section B is h and the height of water column in tube (2) is more than that in tube (1) by $\frac{h}{2}$ distance. If the velocity(m/s) of the fluid at section A is x, then find the value of 6x. (if h = 2.7 m, $g = 10\ m/s^2$)

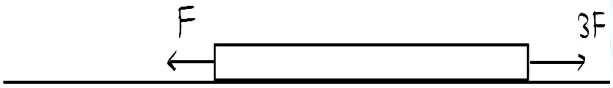


29. Two spherical soap bubbles in vacuum are connected through a narrow tube. Radius of left bubble is R and that of other is slightly smaller than R. Air flows from right to left very slowly. At any instant r_1, A_1, V_1, n_1 are radius, surface area, volume and number of moles of gas in the left bubble and r_2, A_2, V_2, n_2 are same for right bubble. Assume that temperature remains constant:



At some instant number of moles in left bubble is 4 times of number of moles in right bubble then if radius of the left bubble is $r_1 = R\sqrt{\frac{\beta}{5}}$ then find b:

30. A uniform rod of mass m = 2.7 kg length L and area of cross section A is placed on a smooth surface. Forces acting on rod are shown in figure.



Total elastic energy stored in the rod at t = 2 sec is $\frac{\alpha F^2 L}{12AY}$, where Y is Young's modulus of rod. Find α.

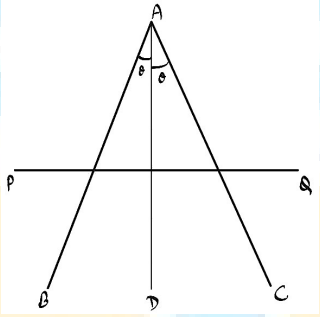
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SECTION - IV (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.
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Paragraph-I for Question Nos. 31 and 32:

In the figure AB, AC and PQ are thin, smooth rigid wires in vertical plane. AB and AC are rigidly joined with Angle bisector being vertical and $\theta = 45^\circ$. Surface tension of liquid is 0.025 N/m which forms a thin film in triangle formed by intersection of wire AB and AC with wire PQ. Wire PQ has 1 g mass and is in equilibrium at given position.



$\left(AB = AC = 30\sqrt{2}\ \text{cm}, g = 10\ \text{m/s}^2\right)$

31. What is the length of wire (in cm) PQ in contact with liquid in equilibrium condition?

32. An additional mass of negligible dimension is placed on wire PQ at mid point of Q. What can be minimum possible value of additional mass (in g) so that it just gets detached from AB and AC?

Paragraph-II for Question Nos. 33 and 34:

A rod having uniform area of cross section of $A = 1\,\text{cm}^2$ is hinged as shown. Mass density of rod increases linearly as a function of distance (x) from hinged end. Length of rod is $l = 3$ m and mars m = 270 g. When rod is submerged in water it remains in equilibrium in horizontal position.



33. If mass density is λ then $\frac{d\lambda}{dx} =$ _____.

34. If water replaced by fluid of specific gravity 1.2 and released from horizontal position then angular acceleration of rod just after release is $\frac{10}{\alpha}\ \text{rad/s}^2$, $\alpha =$ _____

For More Join: @IITJEE_Advanced

CHEMISTRY **MAX.MARKS: 60**

SECTION–I (Maximum Marks: 12)

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Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
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Negative Marks: -1 In all other cases.

35. In BrF_5, considering the effect of one lone pair on **Br**, the maximum number of atoms lying in one plane is x.

Also, among $SiClF_3, SF_6, CCl_4, PCl_5$, find the molecule(s) in which all triangular faces are equilateral.

Choose the correct option:

A) (x) = (4) and molecule (s) = SF_6

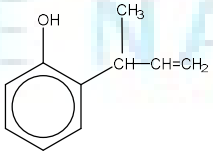
B) (x) = (4) and molecule (s) = SF_6, CCl_4

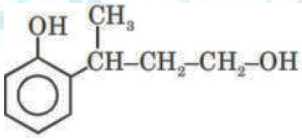
C) (x) = (3) and molecule(s) = SF_6, CCl_4

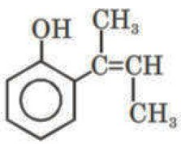
D) (x) = (5) and molecule (s) = CCl_4

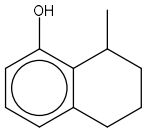
36. An optically active compound A has molecular formula $C_{10}H_{12}O$. A decolourises Bayer's reagent as well as it forms salt with NaOH. A on treatment with HBr forms $B(C_{10}H_{13}OBr)$ as major product which is still chiral. B on treatment with alcoholic solution of KOH gives a compound C which shows stereoisomerism that is also an isomer of A. A on treatment with acidic solution of $KMnO_4$ gives ortho hydroxy benzoic acid as one of the product. A on treatment with B_2H_6 followed by $H_2O_2/NaOH$ produces $D(C_{10}H_{14}O_2)$ which on refluxing with sulphuric acid produced $E(C_{10}H_{12}O)$, E neither decolourises $Br_2 - H_2O$ solution nor forms any salt with NaOH .

Choose the incorrect option:

A) A is 

B) D is 

C) C is 

D) E is 

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37. At 263 K solid compound $A(H_2O)_8$ shows the following equilibria

(Assume 263 K is the sublimation temperature of $H_2O(s)$ under the given conditions):

$A(H_2O)_8(s) \rightleftharpoons A(g) + 8H_2O(g)\ (Kp_1)$

$A(H_2O)_8(s) \rightleftharpoons A(g) + 8H_2O(s)\ (Kp_2)$

At equilibrium, the partial pressures are:

$P(A) = 0.2$ bar

$P(H_2O) = 0.001$ bar

Which of the following is correct?

A) $Kp_1 = 2\times10^{-25}$ and $Kp_2 = 0.2$

B) $Kp_1 = 2\times10^{-25}$ and vapour pressure of ice = 0.002 bar

C) $Kp_1 = 2\times10^{-24}$ and $Kp_2 = 0.2$

D) $Kp_1 = 2\times10^{-25}$ and vapour pressure of ice = 0.01 bar

38. Which of the following reactions produces HCHO ?

A) $CH_4 + O_2 \xrightarrow[\Delta]{Cu,\text{high P}}$

B) $CH_4 + O_2 \xrightarrow[\Delta]{Mo_2O_3}$

C) $CH_3 - CH_3 + O_2 \xrightarrow[\Delta]{(CH_3COO)_2Mn}$

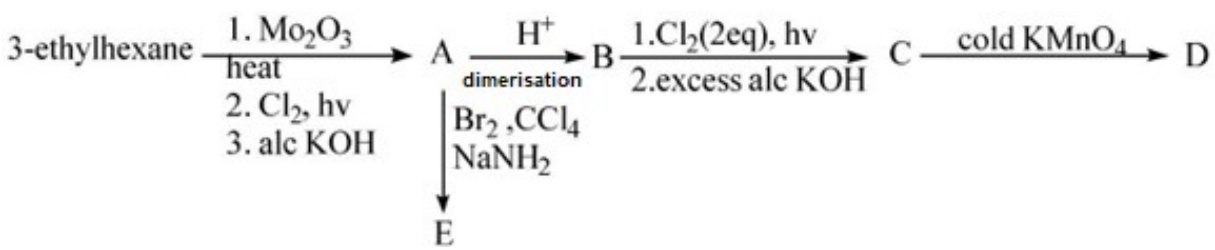
D) $CH_4 + \underset{\text{excess}}{O_2} \xrightarrow{\Delta}$

SECTION - II (Maximum Marks : 12)

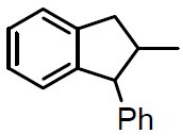
This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

39.



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A) B is 

B) When C reacts with HBr, it forms same TCP and KCP.

C) E can be conveted into A by use of Na/liq NH_3.

D) Number of hyperconjugable hydrogen atoms in B are 2.

40. At temperature T, the following equilibrium is established in a closed container:

$$H_2(g)+I_2(g)\rightleftharpoons 2HI(g)$$

At equilibrium, the total pressure is 2 atm. The volume of the container is suddenly reduced to half isothermally, so that the total pressure becomes 4atm instantaneously.

Which of the following statements is/are correct?

A) The equilibrium shifts in forward direction.

B) Immediately after compression, the reaction quotient Qp remains equal to Kp.

C) Immediately after compression, the partial pressure of each gas becomes double.

D) At equilibrium at the same temperature after compression, the total pressure remains 4 atm.

41. Which of the following statements is(are) correct regarding a catalyst and its effect on equilibrium ?

A) A homogeneous catalyst can change the equilibrium composition of a system.

B) A heterogeneous catalyst does not change the equilibrium composition of the reaction mixture.

C) A catalyst has no effect on the equilibrium constant.

D) A catalyst lowers the activation energy of both the forward and the backward reactions by exactly the same amount.

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SECTION-III (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

42. The reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ is at equilibrium. Now suddenly the volume of the container is doubled. The system is then allowed to attain a new equilibrium state.

How many of the following properties are lesser at the new equilibrium as compared to the old equilibrium?

Assume the entire process to be isothermal.

(a) Moles of N_2O_4

(b) Moles of NO_2

(c) Total number of moles of gas

(d) Concentration of N_2O_4

(e) Concentration of NO_2

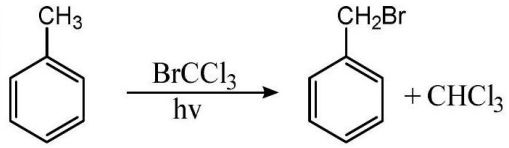
(f) Rate of the forward reaction

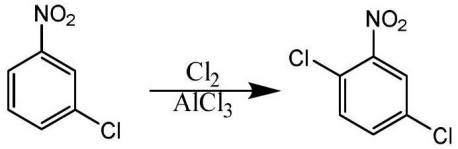
(g) Rate of the backward reaction

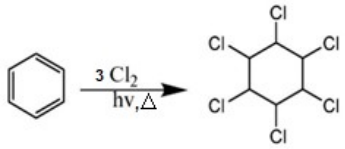
(h) Total pressure

(i) Equilibrium constant

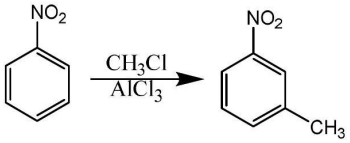
43. Number of reactions which represent correct major product

(A) 

(B) 

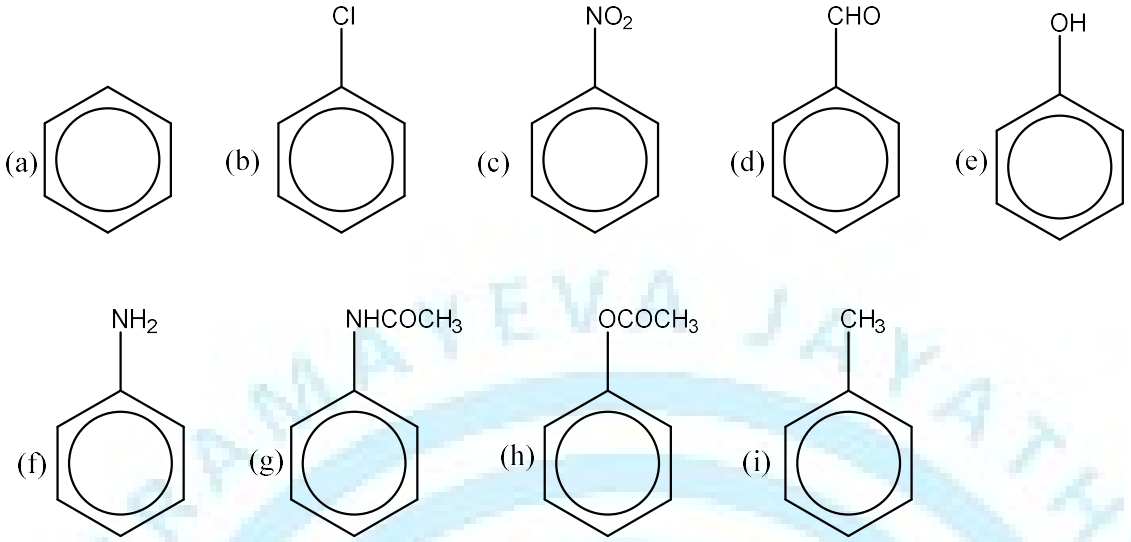
(C) 

(D) 

(E) 

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44. How many of the following compounds can be a suitable substrate to give a good yield of the product in fridel craft's alkylation or acylation reactions ?



45. How many of following are correct statements?

(i) $H_2O > HF > NH_3$ (Boiling point).

(ii) $H_2O_2 > O_2F_2$ (Dipole moment).

(iii) Molecules of type AB_4L are polar & non-planar.

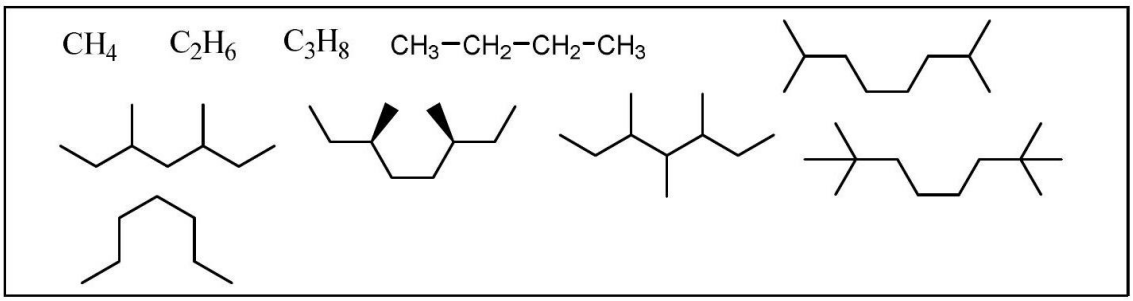
(A = central atom, B = surrounding atom, L = lone pair)

(iv) CO_2 & N_2O have same shape but different polarity.

(v) H-F is the strongest bond amongst all single covalent bonds.

(vi) p_x & d_{xy} orbitals overlap along x-axis to form a p-bond.

46. How many of the following can be prepared in good yield by wurtz reaction



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47. Let N be defined as the sum of the following four results:

(I) Number of pair(s) in which the 1st species has higher % p-character in the orbital used by central atom M for M –H bond formation:

$(CH_4, SiH_4), (NH_3, AsH_3), (H_2O, H_2S), (PH_4^+, PH_3), (CH_4, NH_3), (PH_3, NH_3)$

(II) Number of species (from the list) which contain tetrahedral unit(s):

$N_2O_5(s), PCl_5(s), PBr_5(s), NH_4Cl(s), Na_2SO_4(s), XeF_6(s), Al_2Cl_6$, Pyrosilicate

(III) Number of molecules in which all equatorial M – X bonds are shorter than axial M – X bonds (M = central atom, X = halogen): $PCl_3F_2, PCl_2F_3, PCl_5, SOF_4, ClF_3$

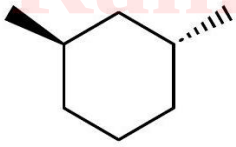
(IV) Total number of orbitals (from the given set) actually used in hybridisation by B in BF_3 and P in PF_5 : $2s, 2p_x, 2p_y, 2p_z, 3s, 3p_x, 3p_y, 3p_z, 3d_{xy}, 3d_{yz}, 3d_{xz}, 3d_{(x^2-y^2)}, 3d_{z^2}$

If the respective answers of (I), (II), (III), (IV) are added to get N, find N.

SECTION - IV (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Paragraph-I for Question Nos. 48 and 49:



$\xrightarrow[hv]{Cl_2}$ all possible monohalogenation products

48. Find the total number of possible mono chloro products (including stereo)

49. Find the total number of Fractions after fractional distillation of products.

Paragraph-II for Question Nos. 50 and 51:

At 25°C, the reaction

$P_2(l) + Q_2(g) \rightleftharpoons 2PQ(g)$

$Kp = 2.5\,atm^{-1}$. The vapour pressure of liquid P_2 at this temperature is 0.3 atm. Initially,

1 mole of Q_2 gas is taken in a closed container.

50. At equilibrium, if the partial pressure of PQ is found to be 0.5 atm, calculate the equilibrium partial pressure(atm) of Q_2.

51. What is the minimum moles of $P_2(l)$ that must be added initially (along with the 1 mole of Q_2 to achieve the given equilibrium state?

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